

1. 【简答题】

Known in the parallel resonant circuit $f_0 = 5 \text{ MHz}$, $C = 50 \text{ nF}$, resonant resistor $R_0 = 100 \Omega$. Calculate Q_0 and L . If the current amplitude of signal source is $I_{sm} = 0.1 \text{ mA}$, calculates U_0 and current values of I_{C0} and I_{L0} .

答案:

$$Q_0 = \omega_0 R_0 C = 2\pi f_0 R_0 C = 50\pi \approx 157$$

$$L = \frac{1}{4\pi^2 f_0^2 C} \approx 20.26 \text{ nH}$$

$$U_0 = I_{sm} R_0 = 10 \text{ mV}$$

$$|I_{L0}| = |I_{C0}| = Q_0 I_{sm} = 15.7 \text{ mA}$$

2. 【简答题】

Series resonant circuit is shown as Fig.1, frequency of the signal source is 1 MHz , voltage amplitude $U_{sm} = 0.1 \text{ V}$, $C = 100 \text{ pF}$. When a short circuit at A-B, the circuit occurs resonance, and the voltage on capacitor C is 10 V . If a capacitive impedance Z_X is connected between A-B, when $C = 200 \text{ pF}$, circuit occurs resonance again, at this time the voltage on capacitor is 2.5 V . Calculates L , Q_0 , Q_L , $B_{0.7}$ and Z_X .

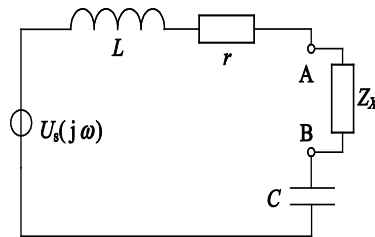


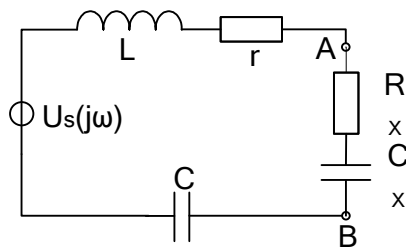
Fig. 1

答案:

$$L = \frac{1}{4\pi^2 f_0^2 C} \approx 253 \mu\text{H}$$

$$Q_0 = \frac{U_{C0}}{U_s} = \frac{10}{0.1} = 100$$

When a capacitive impedance Z_X is connected between A-B,



$$C_{\Sigma} = \frac{C \cdot C_X}{C + C_X} = 100 pF$$

$$\therefore C_X = 200 pF$$

$$U_{C\Sigma} = 5V = Q_L U_S$$

$$\therefore Q_L = 50$$

$$Q_L = \frac{\omega_0 L}{R_{\Sigma}} = \frac{\omega_0 L}{r + R_X}$$

$$\therefore R_X = 15.9 \Omega$$

$$\therefore Z_X = R_X + \frac{1}{j\omega C_X} = (15.9 - j796) \Omega$$

$$B_{0.7} = \frac{f_0}{Q_L} = 20 KHz$$

如果认为电压 2.5V 是所有电容上的电压，那么 $Q_L=25$ ， $B_{0.7}=40KHz$ ， $R_X=47.67$ ohm， $X_Z=-j*796$ ohm 不变，也算对。

3. 【简答题】

In Fig.2, $L = 0.8 \mu H$, $C_1 = 20 pF$, $C_S = 10 pF$, $R_S = 20 k\Omega$, $R_L = 5 k\Omega$, $Q_0 = 100$. If R_L and R_S achieves impedance matching at resonant frequency, what the value of C_2 ? Calculate the resonant frequency f_0 , resonant resistance R_0 , Q_L and $B_{0.7}$.

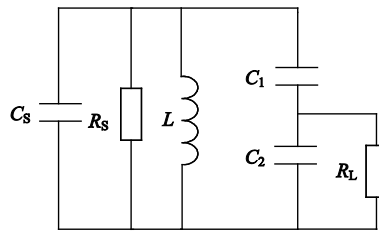


Fig. 2

答案：

$$n = \frac{C_1}{C_1 + C_2}$$

$$R'_L = \frac{R_L}{n^2}$$

Since R_L and R_S achieves impedance matching, so

$$R'_L = R_S, n = 0.5$$

$$C_2 = 20 pF$$

$$C_{\Sigma} = C_s + \frac{C_1 \times C_2}{C_1 + C_2} = 20 pF$$

$$\omega_0 = \frac{1}{\sqrt{LC_{\Sigma}}} = 0.25 \times 10^9 \text{ rad / s}$$

$$f_0 = \frac{\omega_0}{2\pi} \approx 39.8 \text{ MHz}$$

Since Q_0 is finite; so the inductor has a parasitic shunted resistance,

$$R_0 = Q_0 \omega_0 L = 20 k\Omega$$

$$R_{\Sigma} = R_s \parallel R_L' \parallel R_0 = 6.67 k\Omega$$

$$Q_L = \omega_0 R_{\Sigma} C_{\Sigma} = 33.3$$

$$B_{0.7} = \frac{f_0}{Q_L} = 1.2 \text{ MHz}$$

4. 【简答题】

Known that the resonant frequency $f_0 = 10.7 \text{ MHz}$, $C_1 = C_2 = 15 \text{ pF}$, $Q_0 = 100$, $R_L = 10 \text{ k}\Omega$, $R_{in} = 4 \text{ k}\Omega$, $C_{in} = 1 \text{ pF}$, shown as Fig.3. If R_L and R_{in} are impedance matched, calculate N_1/N_2 , L , resonant resistance R_0 , Q_L and $B_{0.7}$.

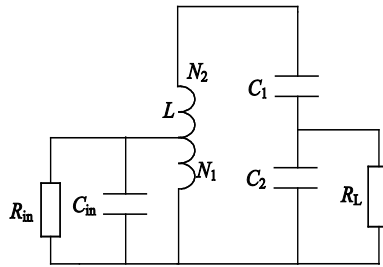


Fig. 3

答案:

$$R_L' = \left(\frac{C_1 + C_2}{C_1} \right)^2 R_L = 400 k\Omega$$

$$R_{in}' = \left(\frac{N_2}{N_1} \right)^2 R_{in} = R_L'$$

$$\frac{N_1}{N_2} = \sqrt{\frac{R_{in}}{R_L'}} = 0.1$$

$$C_X = \frac{C_1 C_2}{C_1 + C_2} = 7.5 pF$$

$$C_{in}' = \left(\frac{N_1}{N_2} \right)^2 C_{in} = 0.01 pF \ll C_X \quad L = \frac{1}{(2\pi f_0)^2 C_X} = 29.53 \mu H$$

$$R_0 = Q_0 / \omega_0 C = 198.42 k\Omega$$

$$R_\Sigma = R_0 // R_s // R'_L = 99.6 k\Omega$$

$$Q_L = \omega_0 R_\Sigma C_x = 50.2$$

$$B = \frac{f_0}{Q_L} = 213 KHz$$